
Contamination with a Location

A Three-Site Epistemic Survey Executed by Two Robots
under Open-RMF, and a Correction to the Floor

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The companion report executes an epistemic policy over forty metres of warehouse and states, at its outset, that floor area and epistemic difficulty are independent quantities. That statement is exact, and it is structural: the proposition under uncertainty carries no argument, one site exists, and the planner is therefore indifferent to geometry. This report removes the independence by giving contamination a location. Exactly one of three sites is contaminated; which one is unknown to every agent. The change is one line of EPDDL. It multiplies the search by four orders of magnitude, doubles the policy, and sends two robots to two aisles. Its principal consequence is epistemic: on one branch the team terminates in possession of the state of a site that no robot approached and no sensor observed.

A second result is negative and concerns the floor of the companion report. The world file carries a saved state block which the simulator applies in preference to the poses the models declare. The two disagree for fourteen models. The navigation graph of the companion report was therefore derived from a building the simulator does not load, and its published figures are superseded by those given here.

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1 Scope

This report concerns a single modification to a previously reported epistemic survey domain, and the consequences of that modification for search, for execution, and for the derivation of the floor the execution takes place on.

The modification is the introduction of a type. Where the companion domain declares the propositional atom *contaminated*, this domain declares the predicate *contaminated(s)* over a type *site*, together with the common-knowledge constraint that exactly one site of three satisfies it. Every other component is held fixed: three agents, an $S5_3$ frame, one semi-private sensing action, two private announcement actions, and a goal of the same form.

Three claims are advanced, and each is bounded in Section 9.

1. A team of agents terminated in possession of knowledge concerning a location that no member visited and no sensor measured, obtained by elimination, with the negative conjunct of the goal intact over that location.
2. Two robots each contributed one measurement, at sites separated by 33.7m on a roadmap derived from the world, with transit costs of 36.5 and 26.8m from their respective chargers.
3. One policy traversed three distinct leaves in three worlds differing by the position of a single object, and satisfied its goal on each.

Four bounds are stated at the outset, since they qualify everything that follows. The nine scripted pedestrians in the world are scenery: they are absent from the navigation graph and are neither perceived nor avoided. The private channel is realised by topic addressing, which establishes that the excluded agent was not addressed and does not establish that it could not have listened. The executor traverses a policy in strict order, so the two robots act sequentially and no claim of concurrency is made. Finally, the corrected floor of Section 4 is established to be a correction; it is not established that no further discrepancy of the same kind remains in the world file.

2 The domain, and the cost of one argument

2.1 Formulation

The domain is stated in EPDDL over a type *site*:

```
(:types site)
(:predicates
  (contaminated ?s - site)
  (on-site ?i - agent ?s - site))
```

The initial state asserts, as common knowledge among the three agents, that exactly one of the three sites is contaminated, and that no agent knows which:

$$K_C \bigvee_{s \in S} \left(\text{contaminated}(s) \wedge \bigwedge_{t \neq s} \neg \text{contaminated}(t) \right), \quad K_C \bigwedge_{i \in \mathcal{A}} \bigwedge_{s \in S} \neg K w_i \text{contaminated}(s).$$

The model therefore carries one world per site, all three designated. The goal requires the scout and the relay each to come to know, for every site, whether that site is contaminated, and requires the observer not to:

$$\bigwedge_{s \in S} K w_{\text{scout}} \text{contaminated}(s) \wedge \bigwedge_{s \in S} K w_{\text{relay}} \text{contaminated}(s) \wedge \bigwedge_{s \in S} \neg K w_{\text{observer}} \text{contaminated}(s).$$

Under the exactly-one constraint the first two conjunctions are satisfied by settling any two sites, since the third is then determined. That implication is the substance of the domain and is exploited by the policy of Section 3.

The goal is written as nine explicit conjuncts. The grounder rejects `:forall` within a goal formula, and the quantified form is therefore expanded at generation time.

2.2 The cost of the argument

Table 1 records the grounded task and the search for the propositional domain and for k sites, $k \in \{2, 3, 4\}$. All rows were grounded with `plank` and solved by `plansys2/EpistemicPlanSolver` under `strategy=aostar` with the `ks` heuristic, three agents throughout, on one machine.

Table 1: Grounded task and search, as a function of the number of sites.

Domain	Atoms	Worlds	Actions	Depth	Expanded	Nodes
propositional	4	2	30	3	40	4
$k = 2$ sites	8	2	48	3	147	4
$k = 3$ sites	12	3	72	6	838 168	8
$k = 4$ sites	16	4	96	not solved within 901 s		

Three observations follow from the table.

The growth is not in the task. The grounded task grows linearly in k : atoms, worlds and ground actions are $4k$, k and $24k$ respectively. The search does not. Expansion counts stand in the ratio 1 : 3.7 : 5 702 across the first three rows, and the plan acquires four further actions between the second row and the third.

Four sites is bounded, not refuted. Under the solver’s default fifteen-second budget the search reports `Timeout at depth 7`; under a budget of nine hundred seconds it reports `Timeout at depth 8` after 901 s and returns no plan. Both outcomes are exhaustion of the budget. No claim is made here that $k = 4$ admits no solution. What is measured is the price of the fourth site: a sixtyfold increase in budget purchased one additional depth iteration.

The three-site search is fast and is nonetheless a hazard. The planner logs the beginning of the search and the solution 6.4 s apart, so 838 168 expansions correspond to approximately 1.3×10^5 expansions per second. That figure is not offered as a property of the search. It is the proximate cause of the first defect of Section 8: the planner and the lifecycle manager of the planning system are served by one executor in one process, and six seconds of uninterrupted search is sufficient to starve the latter.

3 The policy

The solver returns the eight-node conditional plan of Figure 1, at depth 6 with three leaves.

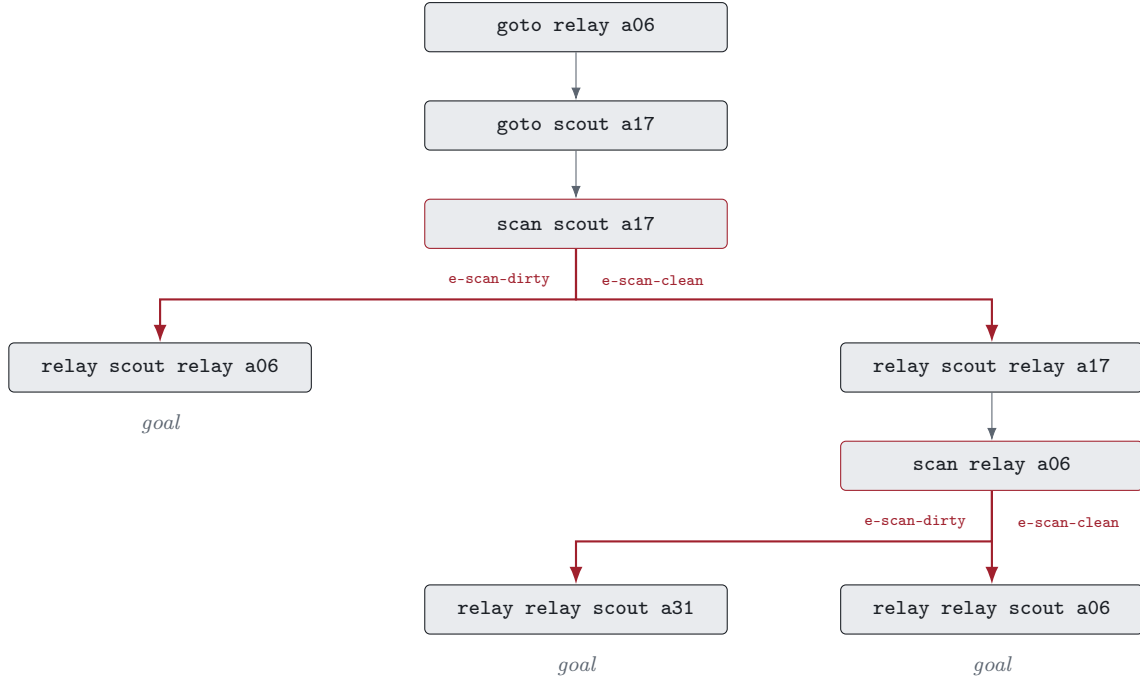


Figure 1: The conditional plan returned for three sites: eight nodes, depth 6, three leaves. Nodes outlined in red are sensing actions and are the only points at which the plan is not a sequence. Branch labels are the observations the branch is taken on.

The assignment of robots to sites is the solver’s. The domain is symmetric in its three sites; the plan of Figure 1 is the plan returned, and it happens to assign the relay a transit of 36.5 m and the scout one of 26.8 m, and to leave a31 unvisited.

Two properties of the plan warrant statement.

3.1 Termination without observation

No action of the plan names a31 as a destination, and no sensing action of the plan is executed at a31. On the leaf reached when both scans return *e-scan-clean*, the exactly-one constraint determines *contaminated(a31)*, and the goal is satisfied for that site by inference alone. The waypoint a31 lies 26.5 m from a17 and 41.4 m from a06. In the corresponding world variant the sensed object stands at a31 for the duration of the run and is never observed.

The propositional domain cannot exhibit a result of this form at any floor area. With one site there is nothing to eliminate, and every proposition the team holds at termination is one that some robot measured.

3.2 Second-order content of the announcements

Every leaf of the plan is a *relay-clean* announcement, including the two that convey which site is contaminated. Consider the leaf reached when the scout’s scan of a17 returns *e-scan-dirty*. The plan there executes (*relay scout relay a06*), whose precondition is $K_{scout} \neg contaminated(a06)$.

The announcement is semantically informative for a reason that is not its content. After the semi-private scan, the relay considers three worlds possible. In the world in which a06 is contaminated, and in the world in which a31 is, the scout’s scan of a17 returned clean and the scout does not know the status of a06; the precondition of the announcement fails at both. The precondition holds at exactly one world, namely the one in which a17 is contaminated. The relay eliminates two worlds from the fact that the scout is in a position to make the utterance, and not from the proposition uttered.

Proposition 1. The information the listener acquires is the restriction of its accessibility relation to those worlds at which the speaker’s knowledge precondition is satisfied. Where that precondition is itself modal, the restriction is not computable from the propositional content of the announcement.

A representation recording only which propositions hold admits no formulation of this step, and a planner reasoning only over such a representation cannot construct it.

4 Correction of the floor

4.1 The state block

A Gazebo world file may carry a `<state>` element recording the pose of every model at the moment the world was saved. The simulator applies that record at load, in preference to the pose declared on the model itself. The world used here, `dynamic_logistics_warehouse`, carries such an element, recording 178 poses against 171 declared models.

Fourteen of the recorded poses differ from the declaration by more than one centimetre. Figure 2 gives the magnitudes.

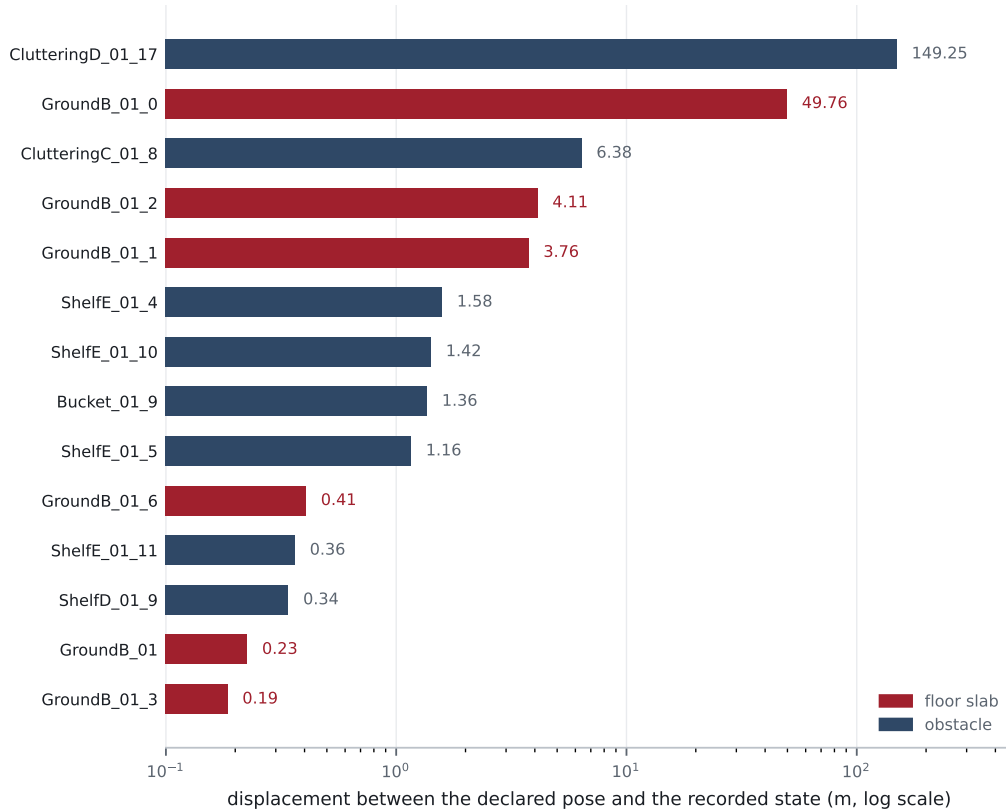


Figure 2: Displacement between the declared pose of a model and the pose recorded in the world’s saved state, for the fourteen models at which the two disagree. Logarithmic axis; the discrepancies span three orders of magnitude. Five of the fourteen are floor slabs, so the correction alters the extent of the floor and not only the placement of obstacles upon it.

The consequence is that a floor plan rasterised from the declarations describes a building the simulator does not load. Obstacles appear where the floor is clear; floor appears where an obstacle stands; and the placement of five of the nine floor slabs is wrong, one of them by 49.76 m. The failure mode this produces in execution is a robot behaving correctly in a world of which the planner holds an incorrect map, which admits no local diagnosis.

4.2 Two subsidiary defects in the same reader

Two further defects were identified in the same component and are reported here because each is silent.

Malformed pose fields. Two of the recorded poses are written with adjacent numeric fields unseparated, in the form 13.8354 41.7933-0.191335. Tokenisation on whitespace yields a field that does not parse as a number. The conventional recovery, a `try / except` returning the origin, places a 14×21 m floor slab at (0,0) and emits no diagnostic. Numeric fields are now extracted by regular expression, which cannot fail in that manner.

Nested models. The world contains a model element named `Untitled` which is a grouping construct and not an asset: it holds a bucket and a desk, each placed relative to the group. Enumeration of the world's direct children returns the group, whose name appears in no footprint table, and both contained obstacles are discarded. The floor beneath them was therefore rasterised as free, and lanes were admitted across it. Model enumeration is now recursive, with poses composed along the nesting.

4.3 Consequences

Table 2 records the derived floor before and after the correction.

Table 2: The derived floor, read from the declared poses and from the poses the simulator loads. The left column comprises the figures published in the companion report; they are superseded by the right.

	Declared poses	Poses loaded
Obstacles	153	155
Waypoints	261	289
Directed lanes	1 652	1 870
Aisle waypoints	35	39
Connected components	1	1

All three chargers are displaced by the correction, one of them by eleven metres. Spawn poses for the fleet were previously held as constants in the launch description and agreed with the graph by inspection. A constant copied from a superseded graph places a robot at coordinates the fleet manager does not associate with it, and the first navigation command is then issued from a position the robot does not occupy. Spawn poses are now read from the charger waypoints of the graph, which removes the possibility of disagreement.

The site of the companion report does not survive the correction: on the corrected floor no side of that waypoint admits the sensed object under the criteria of Section 5. The single-site mission is now configured at `a17`.

5 Selection of sites

5.1 Criteria

A survey site must satisfy three conditions simultaneously, each of which is measurable from the collision extents of the world.

1. **The empty waypoint must read open.** The sensing action reports contamination when the nearest finite laser return falls below a threshold. A waypoint with structure within that threshold reports contamination in every world, including those containing no sensed object, and the sensing action degenerates to a constant.

2. **The sensed object must have an admissible position.** It must stand on floor, clear of the racks, and clear of every lane the fleet may traverse. The third condition is the one the companion report shipped without: an object placed for visual effect overlapped a charger lane by 0.20 m, and a robot was held stationary against it while the fleet reported its task underway, with no component reporting a collision.
3. **The two readings must be separated by the threshold with margin.** A site at which the occupied and empty readings straddle the threshold by centimetres is a site whose branch is determined by the stopping position of the fleet.

These are evaluated from `config/aws_footprints.json`, the same footprint table from which the navigation graph is rasterised, so the two cannot disagree. Thirteen of the thirty-nine aisle waypoints satisfy all three.

5.2 The three sites

Table 3: The three sites, and the admissible position of the sensed object at each. Clearances are computed from the collision extents; the sensed object is a $0.9 \times 0.5 \times 0.9$ m box placed 0.80 m from the waypoint with its major axis directed at it.

Site	Position	Empty	Side	Occupied	Rack	Lane
a17	(1.49, 15.75)	1.48	west	0.35	0.44	0.35
a31	(15.24, 37.00)	1.24	west	0.35	0.87	0.35
a06	(30.24, -1.75)	1.71	east	0.35	0.46	0.35

a17 is a terminal waypoint: one lane enters it and that lane is the only exit. Where a side is admitted at all it is admitted at the minimum lane clearance of 0.35 m, which is the robot radius of 0.25 m together with a margin of 0.10 m.

5.3 Agreement between the model and the sensor

The analytic criteria admit direct comparison against the simulator, and the agreement is not uniform. Two regimes are distinguished.

The predicted reading for the occupied case is reliable. The criteria give 0.35 m for the nearest face of the sensed object, against measurements of 0.32–0.34 m in the companion report and 0.33–0.41 m in the runs of Section 7. The object is a box whose vertical extent the model and the sensor agree upon.

The predicted reading for the empty case is not reliable. The criteria give 1.71 m at a06 where the sensor returned 2.25 m, and 1.48 m at a17 where the sensor returned 0.97 and 1.08 m. Two causes account for the discrepancy in both directions. The criteria are two-dimensional and the sensor is a plane at 0.26 m above the floor, so structure shorter than that plane is an obstacle to the former and invisible to the latter. And the nine pedestrians are excluded from the criteria by construction and are entirely visible to the sensor.

The criteria are therefore serviceable for refusing a site and are not serviceable for predicting a reading. Every admitted site carries in excess of one metre of clearance, and that margin absorbs the half metre of disagreement observed.

6 Execution

6.1 Two sensors

The companion report fits a laser to one robot and records the reason: Gazebo Classic derives the name of a plugin’s node from the plugin, so several robots instantiated from one model description

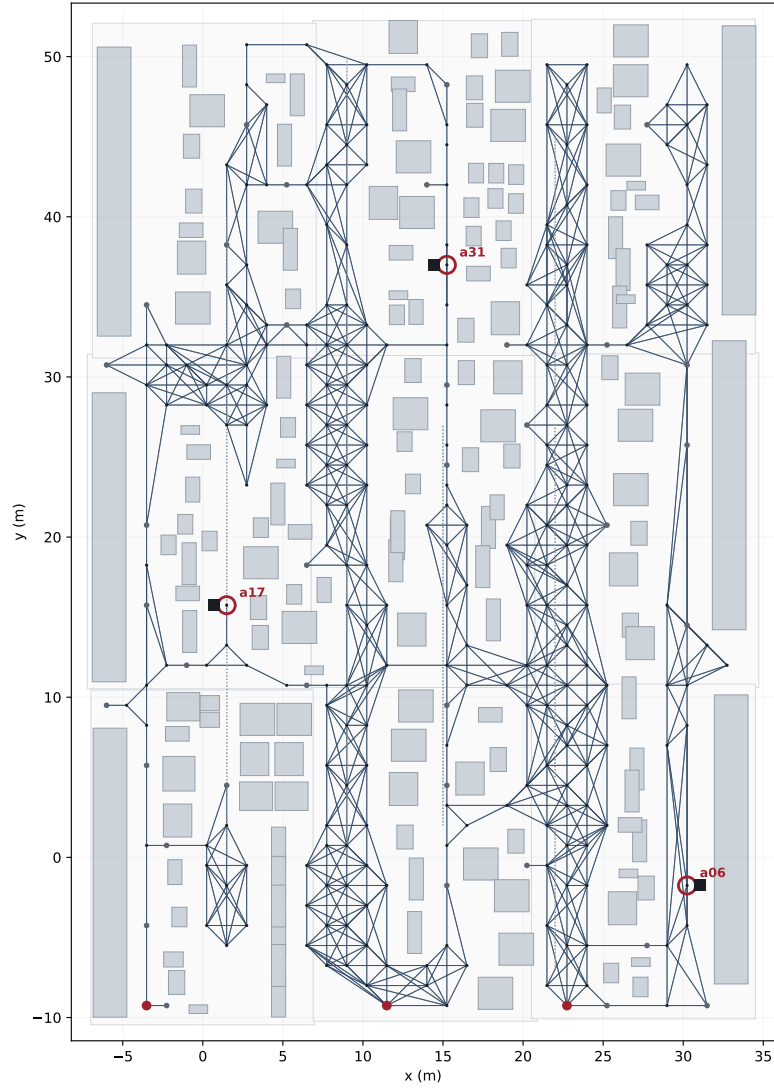


Figure 3: The corrected floor and the roadmap derived from it: 289 waypoints, 1 870 directed lanes, 155 obstacles. Obstacles are shaded; lanes are drawn in outline; chargers are marked in red on the south wall; the three sites are ringed and the position of the sensed object at each is marked by a filled square. The dotted paths are the circuits of the nine scripted pedestrians, which are excluded from the graph.

yield several nodes of one name, and the server terminates. With a single sensing agent the restriction is without cost.

The three-site plan executes sensing actions at two sites with two robots, and the restriction is therefore binding. The ray plugin accepts a ROS namespace, and a namespace is the scope within which a node name is unique. A copy of the laser-bearing model description is generated per sensing robot at launch, with the plugin placed in that robot's namespace; scan topics are consequently `/r1/scan` and `/r2/scan`. Generation at launch is preferred to checked-in copies, which would differ by one string and would diverge upon any modification of the geometry.

6.2 Attribution of observations

The bridge instantiates one action performer per action name, and observations arrive on one topic. A domain with several sites therefore routes every sensing action through one performer and one observation field. The topic is latched, so a performer subscribing after an earlier reading receives that reading immediately.

The consequence for a plan that dispatches two robots to two sites is severe and is not detectable from any single component. The second robot is already positioned at its site when the first reports, so the value available at the second sensing action is a genuine measurement, taken by the correct sensor at the correct place, and attributed to the wrong action. The branch is taken, the product update is applied, the goal is checked, and the value consumed is whichever site reported most recently.

Temporal disambiguation is inadequate, and the inadequacy is instructive. A robot commonly arrives at its site during the `goto` action preceding the scan, so the correct reading is already several seconds old when the sensing action begins. A rule admitting only readings newer than the action discards the correct value and falls through to the ground truth recorded in the task map. Since the launch derives that ground truth from the same argument that positions the sensed object, the fallback is correct, and the defect is therefore invisible in the run.

Observations are consequently published in the form `<place> <outcome>`, and the bridge resolves the entry for the place its own action names, taken from the argument the task map declares as selecting the outcome. Two sensing actions at two places cannot answer for one another under any ordering in time.

6.3 Perception

Both sensing robots carry a planar laser on the mast at 0.33 m, with 180 beams over the full revolution and returns admitted between 0.12 and 3.5 m. The perception node subscribes to the fleet state and to each robot's own scan topic, and withholds a verdict until the fleet has reported that robot within 0.40 m of a site on three consecutive reports. It then reads the nearest finite return, compares it against 0.70 m, and publishes.

The node is not informed of the plan. An observation is generated by a robot arriving at a location in possession of a sensor, and a perception component requiring notification of the action in progress would report the plan back to the system that issued it.

7 The three runs

The floor is instantiated four times. The variants differ in the position of one object and in nothing else: the same hall, the same 155 obstacles, the same nine pedestrians, the same navigation graph, the same binary, the same domain, the same goal, the same threshold of 0.70 m.

Table 4: Three worlds, three leaves. Readings are the nearest finite return reported by the perception node at the point of the corresponding sensing action. Checks comprise nine formulas evaluated against the epistemic state after the fleet came to rest, and two transcripts of address.

Object at	scan a17	scan a06	Leaf	Actions	Checks
a17	dirty, 0.33 m	not reached	relay-clean_scout_relay_a06	4	11 of 11
a06	clean, 1.08 m	dirty, 0.41 m	relay-clean_relay_scout_a31	6	11 of 11
a31	clean, 0.97 m	clean, 2.25 m	relay-clean_relay_scout_a06	6	11 of 11

The fourth variant contains no sensed object. It is a control and is excluded from the table: the domain asserts that exactly one site is contaminated, so a world containing no such object contradicts the problem the planner was given.

The third row is the case for which the domain exists. Both sensing actions return `e-scan-clean`; no measurement is taken at `a31` at any point in the run; and the team terminates in possession of $Kw_{scout} contaminated(a31)$ and $Kw_{relay} contaminated(a31)$.

7.1 The model under product update

Execution is not the marking of actions as complete. Each executed action is an event model $\mathcal{E}$, and its application forms the product $\mathcal{M} \otimes \mathcal{E}$. The state component reports the shape of the result at each step. For the third row of Table 4:

```

applied goto_relay_a06: 3 worlds, 3 designated
applied goto_scout_a17: 3 worlds, 3 designated
applied scan_scout_a17 -> e-scan-clean: 3 worlds, 2 designated
applied relay-clean_scout_relay_a17: 5 worlds, 2 designated
applied scan_relay_a06 -> e-scan-clean: 5 worlds, 1 designated
applied relay-clean_relay_scout_a06: 8 worlds, 1 designated

```

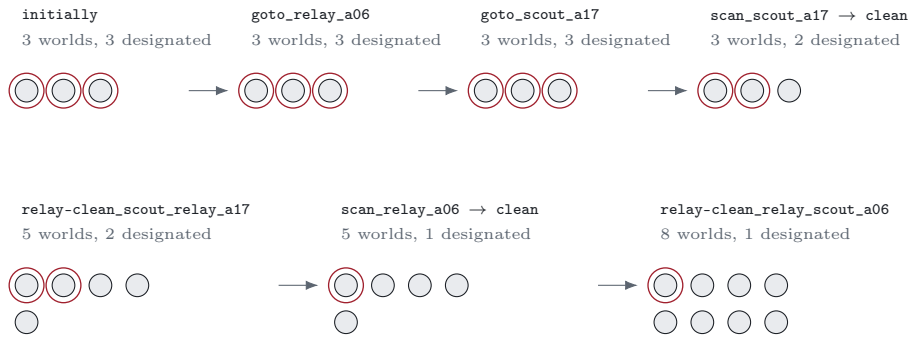


Figure 4: The epistemic model at each product update of the third run. Each disc is a world; a red ring marks a designated world. Counts are those the state component reported. Sensing contracts the designated set; private announcement enlarges the model.

Two directions of movement are visible in the sequence, and both are necessary. Sensing contracts: the first scan reduces the designated set from three worlds to two, and the second from two to one. Private announcement enlarges: the model acquires two worlds at the first announcement and three at the second, because agents outside the audience of an utterance must continue to consider possible a world in which no utterance occurred.

The distinction between the two runs of six actions and the run of four is a consequence of the first observation. A scan returning `e-scan-dirty` settles the question outright, since the contaminated site is then identified; a scan returning `e-scan-clean` eliminates one world of three and leaves the team unable to distinguish the remaining pair, which requires a second sensing action and a second announcement.

7.2 The verdict

Eleven checks are evaluated after the fleet comes to rest. Nine interrogate the epistemic state; two read the transcripts of the agents' radio channels.

```
ok (Kw scout contaminated_a06) holds
ok (Kw scout contaminated_a17) holds
ok (Kw scout contaminated_a31) holds
ok (Kw relay contaminated_a06) holds
ok (Kw relay contaminated_a17) holds
ok (Kw relay contaminated_a31) holds
ok (Kw observer contaminated_a06) does not hold
ok (Kw observer contaminated_a17) does not hold
ok (Kw observer contaminated_a31) does not hold
ok relay was spoken to, 1 time(s)
ok observer was spoken to by nobody
```

The two classes of check are of different evidential strength, and the second is the weaker. A negative epistemic conjunct is satisfied by everything that fails to occur, so a procedure consulting only the model cannot distinguish deliberate exclusion from an absence of communication. The transcript check addresses that gap and does not close it: it establishes that the excluded agent was not addressed.

The transcript check is asserted over the relay alone. The relay is addressed on every branch, the scout's announcement being the first speech act under either outcome of the first scan; the scout is addressed on two branches of three. A check naming both agents would constitute an assertion about which branch was expected, and a run taking the short branch would be reported as a failure for having been short.

8 Defects

Eight defects were identified in bringing the three-site mission into operation. None is an error within a component. Each is a property of the composition, and each was unobservable while the domain carried one site and one sensing robot. Table 5 records them.

Table 5: Defects identified, with their proximate cause and classification.

Symptom	Cause	Class
The navigation graph describes a building the simulator does not load	The world carries a saved state block which Gazebo applies over the declared poses; fourteen disagree, one by 149 m	Geometric
A 14×21 m floor slab is placed at the origin	Two recorded poses have adjacent numeric fields unseparated, so whitespace tokenisation yields a non-numeric field and the conventional recovery returns the origin	Parsing
Lanes are admitted across a bucket and a desk	Both are nested within a grouping model whose name appears in no footprint table; enumeration of direct children discards them	Geometric
A valid policy is returned to a system that has terminated; the mission fails at <code>send_goal failed</code>	The readiness test admitted a configured node, and six seconds of search starved the lifecycle manager sharing the process	Race
A sensing action reports a valid measurement, taken by the correct sensor at the correct place, for the wrong site	One performer per action name and one latched observation topic; the second sensing action receives the first one's value on subscription	Composition
One branch of three does not proceed past <code>Error checking at start reqs</code>	The classical precondition required the speaker of an announcement to have scanned the site it names	Modelling
The executor reports a precondition over a non-existent object, fifty times per second, indefinitely	Four files name the sites and no check compared them; three were renamed and the fourth was not	Duplication
The post-run check reports three formulas <code>UNCHECKED</code> and the mission failed, on a run satisfying its goal	The check component's defaults are the propositional formulas, and this domain grounds one atom per site	Configuration

Three warrant discussion.

8.1 A race the small domain concealed

The planning system is a single process, and a plan request is served on the same executor upon which the lifecycle manager's activation calls arrive. Solving the propositional problem requires forty expansions and returns before the coincidence is observable. Solving the three-site problem occupies the process for six seconds, during which the activation calls go unanswered and the bringup reports `Failed to start plansys2!`. The planner subsequently returns a valid eight-node policy to a system that has already terminated, and the mission fails at `start_plan_execution` with `send_goal failed`, some forty log lines and several seconds removed from the cause.

The readiness test previously admitted any node that answered. A configured node answers, several seconds before the bringup activates anything. The mission now waits for every managed node to report the active state.

The general form is worth recording. Search time is a property of the domain. The bringup is a property of the framework. The two interact because they share a process, and no interface between them expresses the dependency.

8.2 A precondition contradicting the domain

The classical half of the mission is deliberately impoverished: it cannot distinguish a broadcast from an encrypted relay, and it is not intended to. It nonetheless gates the actions the executor drives, and one of its gates encoded an assumption this domain exists to refute. The `relay` action required (`scanned ?from ?s`): the speaker of an announcement must have scanned the site it names. Under the exactly-one constraint the announcement carrying the finding names a site the speaker has not visited, which is the result of Section 3.1.

The defect is observable on one branch of three. Where the first scan returns clean, the announcement names the site just scanned and the precondition holds; where it returns dirty, the announcement names a different site and the executor does not proceed. Two of the three worlds executed to completion with the incorrect precondition in place. The condition is now (`has_scanned ?from`), which requires that the speaker has performed some sensing action.

8.3 A correct refusal

The post-run check is configured by default with the propositional formulas, and this domain has no atom *contaminated*: the grounder produces one atom per site. Presented with formulas naming an atom outside the task, the check reported them `UNCHECKED` and failed the run.

That behaviour is correct and is recorded because the alternative is consequential. A check returning *false* for a formula it cannot evaluate would report a satisfied run as unsatisfied, and, on the negative conjunct, would report an unsatisfied run as satisfied. The distinction between *does not hold* and *could not be evaluated* is load-bearing wherever a goal contains a negative epistemic conjunct.

8.4 One observation without explanation

On the floor as read before the correction of Section 4, the scout halted 2.5 m short of its site, in open floor with 1.05 m of clearance, and did not proceed. The behaviour reproduced across two runs at coordinates agreeing to three decimal places. The robot's own laser reported an object at 0.23 m to its rear, fixed with respect to the world: displacing the robot by 0.86 m displaced the reading by 0.85 m at the same bearing. No model in the world file, at any nesting depth, under either the declared or the recorded poses, and no pedestrian on any leg of its circuit, lies within 1.9 m of that point. The fleet manager reported the robot 10.5 s from its destination for four minutes.

The behaviour has not recurred on the corrected floor and is not explained. It is recorded because a halt presenting as a task permanently underway, with no component reporting an error, is the signature of the class of defect this section is concerned with.

9 What the runs establish, and what they do not

9.1 Established

Each proposition is stated existentially: there exists a run in which it held.

Knowledge without observation. A team terminated in possession of the state of a site that no member visited and no sensor measured. Both sensing actions returned clean, the third site was determined by the exactly-one constraint, and the derived finding was transmitted on the private channel. The negative conjunct held over that site, verified against the model and against the transcript.

Two sensing agents on a derived roadmap. Two robots each contributed one measurement, at sites 33.7 m apart, at transit costs of 36.5 and 26.8 m from their chargers. The selection of sites to inspect, and their order, is determined by the planner, and the sites are separated by transits on a graph derived from the world. This constitutes the coupling between floor and epistemics that the companion report states it does not possess.

Three leaves of one policy. One conditional plan traversed three distinct leaves in three worlds differing by the position of one object, and satisfied its goal on each. A conditional plan reaching its goal on every branch is the object of planning under partial observability, and is not demonstrated by a run traversing one branch.

A floor derived from the poses in use. The correction of Section 4 is established, in the sense that the figures of Table 2 are derived from the poses the simulator applies.

9.2 Not established

Concurrency. The executor traverses a plan in strict order, so the relay’s transit completes before the scout’s begins. Two agents performing physical work is not two agents working simultaneously, and no element of this configuration forces two robots into one corridor at one time. The instrumentation exists; the experiment does not.

Scaling. Three sites is one measurement. Four sites is unsolved within an hour of budget. What is measured is the cost of the third site, and not a curve.

Object recognition. The perception component compares the nearest finite return of a planar scan against a distance selected for these sites. It establishes that an object is proximate to the robot at a waypoint at which, in the empty variant, no object is. It does not identify the object.

Navigation among moving obstacles. The nine pedestrians traverse fixed circuits and are absent from the navigation graph. The fleet neither perceives nor avoids them.

Confidentiality. The private channel is realised by addressing, and any node may subscribe to any topic. The available claim is that the team used a channel on which the excluded agent was not addressed. The stronger claim requires DDS partitions or SROS 2 permissions, which would be constructed from these same declared audiences; what separates the two is enforcement.

Completeness of the floor correction. It is established that the declared poses are not the poses applied. It is not established that no further discrepancy of the same kind remains, and one halt on the superseded floor is unexplained.

Reproduction

```
bash tools/make_site_worlds.sh /tmp
ros2 launch warehouse_xl_rmf_demo survey_sites_launch.py \
  contaminated:=a31 world:=/tmp/warehouse_xl_a31.world

python3 tools/check_sites.py # every admissible site
python3 tools/plot_state_drift.py --out drift.pdf # the pose disagreement
```

`contaminated:=` selects the variant, and admits `a17`, `a31`, `a06` and `none`. `plan_only:=true` `rmf:=false` returns the plan without a simulator. Sites are read from the EPDDL problem by the mission component and are checked against the launch description at startup, so the four files naming them cannot diverge without refusal.